Exercise Sheet 5

Throughout, $\int_G f(g)\,dg$ is the normalised Haar integral of Lecture 5. Inner products are antilinear in the first slot. Exercises 5.1–5.6 are the core work. Exercise 5.7 is an optional extension.

Exercise 5.1. The circle and its finite subgroups. Let $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ with

$$\int_{\mathrm{U}(1)} f(z)\,dz = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta.$$

Fix $N \geq 2$, put $\zeta = e^{2\pi i/N}$ and $\mu_N = \{\zeta^k : 0 \leq k < N\}$. For a function f on $\mathrm{U}(1)$, let

$$A_N(f) = \frac{1}{N}\sum_{k=0}^{N-1} f(\zeta^k).$$

(a) For $n \in \mathbb{Z}$, compute $\int_{\mathrm{U}(1)} z^n\,dz$ and $A_N(z^n)$.

(b) Let $f(z) = \sum_{|n|\leq M} c_n z^n$ with $M < N$. Show that $A_N(f) = \int_{\mathrm{U}(1)} f(z)\,dz$. Give an f with $M = N$ for which the two values differ.

(c) Let $\Pi(z) = \mathrm{diag}(z^N, z)$ on $\mathbb{C}^2$. Compute $P = \int_{\mathrm{U}(1)} \Pi(z)\,dz$ and $P_N = \frac{1}{N}\sum_k \Pi(\zeta^k)$. Identify their images.

Exercise 5.2. Moments on SU(2). Write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2), \qquad a = \cos\eta\, e^{i\xi_1}, \quad b = \sin\eta\, e^{i\xi_2},$$

with $0 \leq \eta \leq \pi/2$ and $0 \leq \xi_1, \xi_2 \leq 2\pi$. Use the Haar integral

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\int_0^{2\pi}\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2.$$

Let $p, q, r, s \geq 0$ be integers.

(a) Show that $\int_{\mathrm{SU}(2)} a^p \bar{a}^q b^r \bar{b}^s\,dg = 0$ unless $p = q$ and $r = s$.

(b) Show that

$$\int_{\mathrm{SU}(2)} |a|^{2p}|b|^{2r}\,dg = \frac{p!\,r!}{(p+r+1)!}.$$

(c) Deduce the integrals of $|a|^2$, $|a|^4$, $|b|^4$ and $|a|^2|b|^2$. Check them against $|a|^2 + |b|^2 = 1$.

Exercise 5.3. Coefficient norms in V_2. Let $g \in \mathrm{SU}(2)$ be written as

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

You may use these facts. A monomial $a^p\bar{a}^q b^r \bar{b}^s$ has Haar integral 0 unless $p = q$ and $r = s$. Moreover $\int |a|^2 = \int |b|^2 = \frac{1}{2}$, $\int |a|^4 = \int |b|^4 = \frac{1}{3}$ and $\int |a|^2|b|^2 = \frac{1}{6}$. Let e_1, e_2 be the standard basis of $\mathbb{C}^2$, and write $e_{jk} = e_j \otimes e_k$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$ with the tensor-product inner product. Let V_2 be spanned by

$$f_0 = e_{11}, \qquad f_1 = \tfrac{1}{\sqrt{2}}(e_{12} + e_{21}), \qquad f_2 = e_{22},$$

with $\Pi(g) = g \otimes g$ restricted to V_2. You may assume that V_2 is invariant and irreducible of dimension 3. Put $D_{ij}(g) = \langle f_i, \Pi(g) f_j\rangle$.

(a) Show that f_0, f_1, f_2 are orthonormal. Compute $\Pi(g)f_0$ and $\Pi(g)f_1$ in this basis.
(b) Write down the first two columns of $D(g)$.
(c) Verify $\int_{\mathrm{SU}(2)} |D_{ij}|^2\, dg = \tfrac{1}{3}$ for $j = 0, 1$ and $i = 0, 1, 2$.
(d) Verify $\int \overline{D_{00}}\, D_{11}\, dg = 0$ and $\int D_{11}\, dg = 0$. Say which case of Schur orthogonality each identity illustrates.

Exercise 5.4. The invariant-vector projector on matrices. Let $\mathrm{SU}(2)$ act on $M_2(\mathbb{C})$ by $\mathrm{Ad}(g)X = gXg^{-1}$. Use the inner product $\langle X, Y\rangle = \operatorname{tr}(X^*Y)$, and let E_{jk} be the matrix units. Write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = g^* = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

You may use the orthogonality relation for $V_1 = \mathbb{C}^2$, namely $\int \overline{g_{kl}}\, g_{ij}\, dg = \tfrac{1}{2}\delta_{ik}\delta_{jl}$. You may also use $\int |a|^2 = \int |b|^2 = \tfrac{1}{2}$ and the rule that $a^p \bar{a}^q b^r \bar{b}^s$ integrates to 0 unless $p = q$ and $r = s$.

(a) Show that the E_{jk} are orthonormal and that each $\mathrm{Ad}(g)$ is unitary.
(b) Compute $P(X) = \int_{\mathrm{SU}(2)} gXg^{-1}\, dg$ for every X.
(c) Check (b) directly for $X = E_{11}$ and $X = E_{12}$.
(d) Identify the invariant matrices. Confirm that P is the orthogonal projection onto them.

Exercise 5.5. Averaging a non-invariant inner product. Let $\mathrm{U}(1)$ carry the Haar integral of Exercise 5.1, so $\int z^n\, dz = \delta_{n0}$. Put

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \Pi(z) = S \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix} S^{-1} \quad \text{on } \mathbb{C}^2,$$

with the standard inner product $\langle v, w\rangle_0 = v^*w$.

(a) Compute $\Pi(z)$. Show that $\mathbb{C}e_1$ is invariant but its standard complement $\mathbb{C}e_2$ is not.
(b) Compute $H = \int \Pi(z)^*\Pi(z)\, dz$, so that the averaged inner product is $\langle v, w\rangle = v^*Hw$. Show that H is positive definite.
(c) Verify $\Pi(z)^*H\Pi(z) = H$ by multiplying matrices.
(d) Find the H-orthogonal complement of $\mathbb{C}e_1$ and check that it is invariant.
(e) Which property of the Haar integral guarantees (c) without computation?

Exercise 5.6. Inversion invariance and characters. Let G be a compact Lie group. You may use left and right invariance, positivity and normalisation of the Haar integral. You may also use the cited uniqueness theorem: a linear functional J on $C(G)$ with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all h is the Haar integral.

(a) Let $J(f) = \int_G f(g^{-1})\, dg$. Show that J satisfies the hypotheses of the uniqueness theorem. Conclude that $\int_G f(g^{-1})\, dg = \int_G f(g)\, dg$.
(b) Let Π be a unitary representation with character $\chi_\Pi(g) = \operatorname{tr}\Pi(g)$. Show that $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$.
(c) Let Π, σ be irreducible unitary representations. Using the character orthonormality of Lecture 5, show that $\int_G \chi_\Pi(g)\chi_\sigma(g^{-1})\, dg$ equals 1 if $\Pi \cong \sigma$ and 0 otherwise.

Exercise 5.7. Optional extension: where compactness enters. Let

$$G = \left\{ g(\alpha,\beta) = \begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} : \alpha > 0,\ \beta \in \mathbb{R} \right\}.$$

This is a noncompact Lie group with global chart (α, β). Its Lie algebra has basis $X = E_{11}$ and $Y = E_{12}$. A continuous density δ on G is given by its coefficient $\rho(\alpha, \beta)$ in this chart. If a diffeomorphism F is written in coordinates as $u \mapsto F(u)$, then $F^*\delta$ has coefficient $u \mapsto \rho(F(u))\,|\det DF(u)|$.

(a) Show that $g(\alpha,\beta)\,g(\alpha',\beta') = g(\alpha\alpha', \alpha\beta' + \beta)$. Compute $\mathrm{Ad}(g)X$, $\mathrm{Ad}(g)Y$ and $j(g) = |\det \mathrm{Ad}(g)|$.

(b) Show that $\rho_L = \alpha^{-2}$ is invariant under all L_h, and $\rho_R = \alpha^{-1}$ is invariant under all R_h.

(c) Show that R_h^* multiplies ρ_L by $j(h)^{-1}$. Which step of the right-invariance proof in Lecture 5 fails here?